# Uncovering HLA loss of heterozygosity in cancer for the improvement of personalized neoTCR immunotherapy with PACT-ESCAPE

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## Introduction

- PACT Pharma is a clinical-stage company (NCT03970382) that develops personalized neoTCR-T cell therapies using a state-of-the-art approach to discover and validate predicted neoepitopes and their cognate T cell receptors.
- Genetic loss of human leukocyte antigen (HLA) alleles and alterations in allelic expression are important mechanisms of resistance to immunotherapy. Targeting neoantigens presented by deleted, mutated, or downregulated HLA alleles is likely to have reduced efficacy due to the lack of epitope presentation on the tumor surface.
- PACT-ESCAPE is a method that integrates DNA and RNA sequencing data to quantify HLA LOH and allelic imbalance at HLA loci, as well as discover deleterious somatic variants in antigen presentation genes.

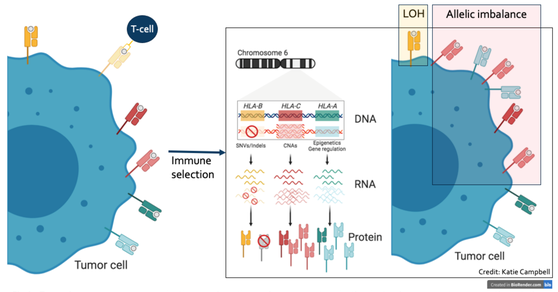



**Fig 1.** Tumor immune escape occurs by three main pathways in response to selection by the immune system: somatic mutation, copy number alteration, and downregulation of antigen presentation genes. Allelic imbalance at HLA loci, and its most extreme form, loss of heterozygosity (HLA LOH), occurs when copy number evolution in the tumor results in a difference in the number and/or expression of HLA alleles at a locus, impacting the density of epitopes on the tumor surface for the immune system to recognize.

## Methods

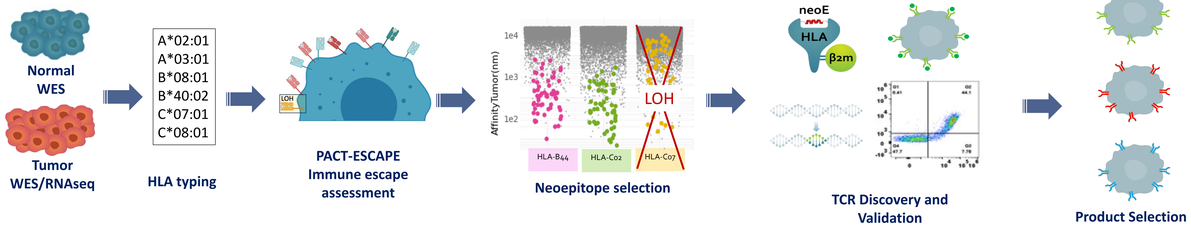
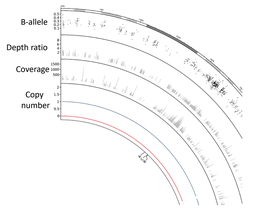



**Fig 2. Overview of the PACT process for selecting neoepitopes and validating TCRs from patient samples**. DNA and RNA sequencing is used to identify alleles impacted by immune escape in the tumor and exclude them from neoepitope selection. TCRs against efficacious epitopes are then retrieved from the patient using PACT's comPACT and imPACT technology, gene edited, and validated for product selection and delivery to the patient. See poster **5167** for more details.

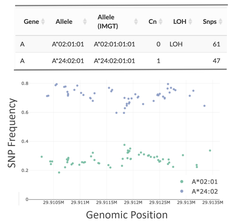
### PACT-ESCAPE enables detection of immune escape using DNA and RNA sequencing

#### 1. Genome-wide copy number



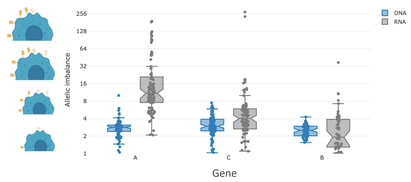

Copy number alterations are assessed across the tumor genome using an algorithm optimized for the highly polymorphic HLA loci. HLA copy number is stabilized by integrating flanking regions of the genome and tumor purity/ploidy calculated to derive absolute copy number. Copy number for other genes involved in antigen presentation machinery (e.g.B2M) are also reported to screen patients unlikely to respond to therapy.

#### 2. HLA allelic copy number



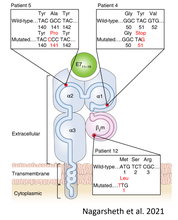

High-resolution HLA typing enables identification of allele-specific SNPs, which are evaluated statistically to confidently assign the patient's lost and kept HLA allele at loci affected by LOH. Neoepitopes cognate with the lost allele can then be removed before TCR discovery, while retaining epitopes presented by the kept allele.

#### 3. RNA/DNA allelic imbalance at 26 HLA loci




Allelic imbalance is the relative abundance of allele 1 vs 2 at a locus, quantitated for 26 HLA loci across chromosome 6p using RNA and DNA sequencing. For RNA, an allelic imbalance of 1 indicates that the expression of each allele is equal, while values greater than 1 indicate expression of one allele is higher than the other, the identity of which is identified by allele-specific SNPs. DNA allelic imbalance is similarly interpreted and represents allelic copy number differences in the tumor.

#### 4. Somatic variant discovery




Nagarsheth et al. 2021

Variant detection in the polymorphic HLA class I genes is performed using HLA typing, allowing exclusion of neoepitopes presented by mutated HLA alleles. Precision detection of deleterious somatic mutations in other antigen presentation genes also identify patients that are unlikely to respond to TCR therapy.

## Results

### PACT-ESCAPE HLA LOH calls are concordant with a state-of-the-art machine learning algorithm

PACT-ESCAPE calls were benchmarked against DASH, a validated DNA-based HLA LOH machine learning classifier (*Pyke et al.* in press). Seventeen patients from six cancer types were sequenced on the same sequencing platform and assessed. 43/44 loci had concordant DNA HLA LOH calls at HLA-A,B, and C. The discordant call was reported as a loss of C*07:02 in a melanoma patient (0612) by PACT-ESCAPE, but not in DASH.

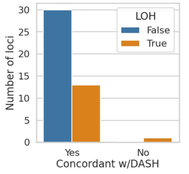



**Fig 3.** PACT-ESCAPE HLA LOH concordance with DASH.

### PACT-ESCAPE DNA and RNA analysis improves confidence in HLA LOH assessment

PACT-ESCAPE's LOH call in 0612 was supported by a long segment (15MB), with LOH (red) spanning most of the HLA region. The minor allele frequency (BAF), used for allelic copy number, was well below the threshold for LOH in the HLA region (dashed gre

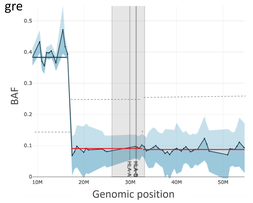



**Fig 4.** B-allele frequency across chr6p for patient 0612. Red line is the LOH segment spanning the HLA region (grey shading), black line and blue shading is the BAF and interquartile across 1Mb windows. Dashed grey line is the LOH threshold.

DNA and RNA allelic imbalance were both high for C*07:01/C*07:02, supporting the LOH call. DNA and RNA imbalance are the relative DNA abundance and RNA expression between alleles at a locus.

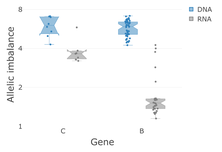



**Fig 6.** Allelic imbalance of SNPs at heterozygous HLA loci for 0612. A value of one indicate no allelic imbalance.

The number of SNPs separating C*07:01/C*07:02 was small (n=8), however their difference in SNP frequency, supported by PACT-ESCAPE's statistical analysis (p=0.015), designates C*07:02 as the lost allele.

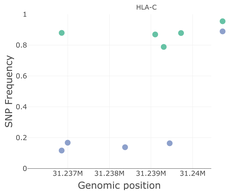



**Fig 5.** Normalized SNP frequency for 0612 C*07:01 and C*07:02. When SNP frequencies differ between alleles, the allele with the lower SNP frequency can be assigned to the lower copy number state.

DNA allelic imbalance across 26 HLA loci matches the expected value for LOH (top line), in agreement with copy number result in Figure 4. Confidence in LOH increases with tumor purity, which was high (0.71) in patient 0612.

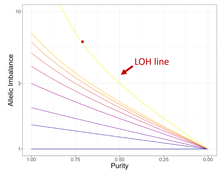



**Fig 7.** DNA allelic imbalance (AI) across 26 HLA loci for patient 0612. Lines are the expected imbalance for different tumor purities. LOH is the top (yellow) line.

### Pan-cancer cohort reveals HLA LOH is a common escape mechanism

Loss of heterozygosity was assessed at HLA class I loci in 122 patients across 8 tumor types recruited from PACT's clinical trial (NCT03970382). Patients had been treated with at least one line of therapy and were predominately metastatic at screening.

HLA LOH was observed in **19.6%** of patients, with the highest frequency in ovarian cancer and the lowest in breast cancer.

HLA LOH almost always resulted from a genomic deletion spanning all three HLA class I alleles. Of the patients with HLA-LOH that were heterozygous at HLA-A,B, and C, 86% had HLA LOH at all three loci.

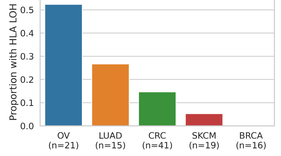



**Fig 8.** HLA LOH frequency across tumor types with greater than 10 patients.

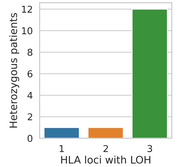
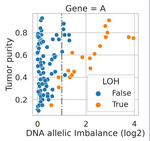



**Fig 9.** Number of HLA loci with LOH in patients heterozygous at HLA-A,B and C.

### Differential expression of RNA alleles occurs in patients with and without HLA LOH

HLA LOH was associated with higher allelic imbalance (AI) in DNA and RNA, as expected if the loss of one allele reduces its DNA abundance and RNA expression (Q1). Some patients also showed AI in RNA but not DNA (Q2), indicating differential regulation of HLA alleles, and potentially, differences in epitope presentation on the tumor surface. Three patients had LOH but low AI (Q3), an artifact of low tumor purity attenuating AI signal (right plot). A fourth group had low RNA AI but high DNA AI (Q4), especially for HLA-B, where RNA AI did not track DNA AI with the same fidelity as HLA-A and HLA-C.



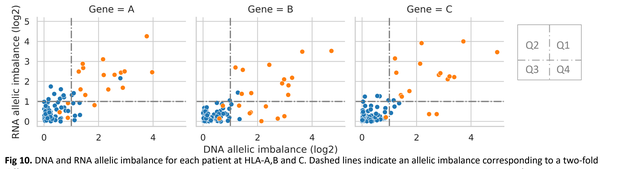




**Fig 10.** DNA and RNA allelic imbalance for each patient at HLA-A,B and C. Dashed lines indicate an allelic imbalance corresponding to a two-fold difference in DNA abundance or RNA expression of one allele over the other. Top right: Tumor purity and DNA imbalance for each patient at HLA-A.

### Detection of deleterious somatic mutations

Two patients, 0908 (HNSC) and 0325 (melanoma), exhibited deleterious mutations in HLA-A (p.Q165C*) and B2M (p.Q28fs) respectively. In PACT's neoepitope prioritization pipeline, allele-specific HLA mutations can be mitigated by selecting neoepitopes for unaffected alleles (Fig 1). Deleterious mutations in B2M affect the presentation of all alleles and thus can result in poor outcomes for immunotherapy (*Jhunjhunwala* et. al 2021).

## Conclusions

- PACT-ESCAPE provides evidence for HLA LOH using both DNA and RNA sequencing, quantifies differential expression of HLA alleles, and detects somatic mutations affecting antigen presentation.
- HLA LOH was found in 19.6% of PACT's clinical cohort, varied extensively among tumor types, and most often spanned all three HLA-A,B, and C loci.
- Immune escape reduces the efficacy of immunotherapy, and thus detection of HLA LOH and other alterations in antigen presentation genes is a key step in selection of T-cell therapies.

**References**

- Jhunjhunwala, S., et al. (2021). Antigen presentation in cancer: Insights into tumour immunogenicity and immune evasion. *Nature Reviews Cancer*, *21*(5), 298–312
- Nagarsheth, N. B. et al. (2021). TCR-engineered T cells targeting E7 for patients with metastatic HPV-associated epithelial cancers. *Nature Medicine*.
- Pyke, R. M. et al. in press. Validated machine learning algorithm with sub-clonal sensitivity reveals widespread pan-cancer human leukocyte antigen loss of heterozygosity. Nature Communications.